2. MATLAB matrices and their elements

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Recitation due Jan 24, 2024 20:30 IST

Accessing elements of a matrix

▶ 2:20 / 2:20 ▸ 2.0x

3 element vector of our column indices, which we enclose in parentheses.

And look at that!

The result is a two by three matrix containing exactly the subset of data we wanted.

When analyzing data, you might find yourself needing to extract an entire row or column.

Extracting a whole row or column of data is so common, there's a shortcut.

A single colon references all of the elements in that dimension together.

That's better.

Here, the colon in the row dimension indicates that we want all of the rows

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Entries of a matrix (External resource) (1.0 points possible)

Entries of a matrix

1. Suppose we have created a matrix $\mathbf{A}$. Then we can extract the element of the mat defining it as a variable a:

```
a = A(m,n)
```

For example, if $\mathbf{A}$ is the following matrix

$$\mathbf{A} = \begin{bmatrix} -1 & 1 \\ 3 & 2 \end{bmatrix}$$

then the command

```
a = A(2,1)
```

will define a variable a with the value 3.

2. We can also extract the columns (or rows) of a matrix. The command

```
v = A(:,1);
```

will create a column vector whose elements are the same as those in the first column o obtain

$$\mathbf{v} = \begin{bmatrix} -1 \\ 3 \end{bmatrix}$$

Now do the following. Use the script below to create a 3×3 matrix $\mathbf{B}$ whose entries ar

```
B = randi([0,50],3);
```

Then:

1. Define a new variable b as the element in the 2nd row and 3rd colum
2. Define a new row vector $\mathbf{v}$ which corresponds to the second row of $\mathbf{B}$

Calculator

2. Define a new row vector v which corresponds to the second row of B.

You should use the same syntax explained above. You will not get full credit for simply t

Script

Save Reset

MATLAB Documentation (https://www.mathworks.com/help/)

```
% Firstly enter a 3x3 matrix B of random integers
B = randi([0,50],3);
% Now define b and v
b = B(2,3);
v = B(2,:);
```

Run Script

Assessment: All Tests Passed

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Value of b

Value of v

Output

```
Code ran without output.
```


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